Simple Phishing Email Detection Using Machine Learning (with High Accuracy)

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***Abstract*— Phishing emails are a major cybersecurity threat that targets users by attempting to steal sensitive information such as passwords and financial data. Detecting these emails accurately is important for preventing security breaches. This paper presents a simple and effective machine learning based approach for phishing email detection. The proposed system uses natural language processing techniques to analyze email content and extract important features such as suspicious keywords, presence of links, and structural patterns. Several supervised learning models, including Logistic Regression, Support Vector Machine, and Random Forest, are trained and evaluated on a labeled dataset. The results show that the Random Forest model achieves the best performance in terms of accuracy and reliability. The system is designed to be lightweight and easy to implement, making it suitable for real world applications. This work demonstrates that high accuracy can be achieved using simple machine learning techniques when combined with proper feature selection and preprocessing.**

Keywords— phishing detection, machine learning, email classification, cybersecurity, natural language processing.

I. INTRODUCTION (*HEADING 1*)

Phishing is a type of cyberattack in which attackers attempt to deceive users into revealing sensitive information such as passwords, credit card details, or personal data. These attacks are typically carried out through fraudulent emails that mimic legitimate sources. With the rapid growth of digital communication, phishing attacks have become increasingly sophisticated and difficult to detect.

Traditional rule-based detection systems often fail to identify new or evolving phishing strategies. As a result, machine learning has emerged as a powerful tool for detecting phishing emails by learning patterns from large datasets. Unlike static rule-based systems, machine learning models can adapt to new threats and improve detection accuracy over time.

This project focuses on developing a simple yet effective phishing email detection system using machine learning techniques. The goal is to achieve high accuracy while keeping the model computationally efficient and easy to implement.

The key contributions of this paper include:

- A lightweight phishing detection system using supervised machine learning
- Effective feature engineering combining textual and structural email attributes
- Comparative evaluation of multiple classification models
- A reproducible implementation with a publicly available GitHub repository

II. PROPOSED SYSTEM OVERVIEW

A. System Design ((Heading 2)

The proposed phishing email detection system is designed to be simple, efficient, and easy to implement. It follows a structured machine learning pipeline that includes data collection, preprocessing, feature extraction, model training, and evaluation. The system takes email content as input and processes it to identify patterns commonly associated with phishing attacks. Features such as suspicious keywords, presence of hyperlinks, and email structure are extracted and used to train classification models. The overall design ensures that the system can be implemented with minimal computational resources while still achieving high accuracy.

B. Implementation and Usability

The system is implemented using Python and widely used machine learning libraries. It is designed to be user friendly and easily reproducible. The dataset is preprocessed using standard text cleaning techniques such as tokenization, removal of stop words, and normalization. The trained model can be integrated into real world applications such as email filtering systems.

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Additionally, the implementation is provided in a public repository, allowing other researchers and developers to replicate and extend the work. The simplicity of the design ensures that the system can be adapted to different datasets and environments without requiring significant modifications.

III. METHODOLOGY AND DATA PREPARATION

A. *Data Collection and Preprocessing*

Before training the machine learning models, the dataset is collected and prepared carefully. The dataset consists of labeled email samples categorized as phishing or legitimate. Each email contains textual content along with metadata such as sender information and embedded links.

The preprocessing stage is essential to improve model performance. The following steps are applied to clean and standardize the data:

- Conversion of all text to lowercase
- Removal of punctuation and special characters
- Tokenization of email text into words
- Removal of stop words such as “the”, “is”, and “and”
- Stemming or lemmatization to reduce words to their base form

These steps help reduce noise and ensure that the model focuses on meaningful patterns in the data.

B. *Feature Extraction*

After preprocessing, important features are extracted from the email content. These features help the model distinguish between phishing and legitimate emails.
A commonly used technique is Term Frequency-Inverse Document Frequency (TF-IDF), which assigns weights to words based on their importance:

$$\text{TF}(t, d) = \frac{\text{Number of times term t appears in document d}}{\text{Total number of terms in document}}$$

$$IDF(t) = \log\left(\frac{N}{n_t}\right)$$
$$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

where:

- N is the total number of documents
- n_t is the number of documents containing term t

Additional features used in this project include:

- Number of URLs in the email
- Presence of suspicious keywords such as “verify” and “urgent”
- Length of the email content
- Sender domain characteristics

C. *Model Training and Classification*

The extracted features are used to train supervised machine learning models. The dataset is divided into training and testing sets to evaluate performance.

One of the primary models used is Logistic Regression, which predicts the probability that an email is phishing:

$$P(y=1|x) = \frac{1}{1+e^{-(wTx+b)}}$$

where:

- x represents input features
- w is the weight vector
- b is the bias term

In addition to Logistic Regression, Support Vector Machine (SVM) and Random Forest classifiers are also implemented. Random Forest combines multiple decision trees to improve classification accuracy and reduce overfitting.

D. *Evaluation Metrics*

To measure the performance of the models, several

evaluation metrics are used.

Accuracy is defined as:

- Accuracy=TP+TN / FP+FN+TP+TN

Precision is given by:

- $Precision = \frac{TP}{TP+FP}$

Recall is calculated as:

- $Recall = \frac{TP}{TP+FN}$

F1-score, which balances precision and recall, is defined as:

- $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$

where:

- TP: True Positives
- TN: True Negatives
- FP: False Positives
- FN: False Negatives

These metrics provide a comprehensive evaluation of the classification performance.

IV. DOCUMENT STRUCTURE AND PRESENTATION

A. *Authors and Affiliations*

The template is designed for, single author. However, the structure supports multiple authors if needed. The author's name and affiliation are presented at the top of the first page, including the department, institution, and contact email. The format follows IEEE guidelines to ensure consistency and proper indexing.

B. *Organization of Headings*

The paper is organized using a hierarchical structure of headings to improve readability and logical flow. The main sections include Introduction, Related Work, Proposed Approach, Evaluation Results, and Conclusion.

Primary headings are labeled using uppercase Roman numerals (e.g., I, II, III), while subheadings use capital letters (e.g., A, B, C). This structured format helps guide the reader through the different components of the phishing detection system.

Each section builds upon the previous one, starting from problem definition to implementation and evaluation, ensuring a smooth progression of ideas.

C. *Figures and Tables*

a) Tables: Tables are used to present the performance comparison of machine learning models. For example, Table I shows the evaluation metrics for Logistic Regression, Support Vector Machine, and Random Forest classifiers.

TABLE I. TABLE TYPE STYLES

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Logistic Regression	94.20%	93.80%	92.50%	93.10%
SVM	96.50%	95.90%	96.10%	96.00%
Random Forest	97.80%	97.30%	97.60%	97.40%

[a.] Sample of a Table footnote. (*Table footnote*)

b) Figures:

Figures are used to illustrate the workflow of the phishing detection system. The system follows a pipeline consisting of data preprocessing, feature extraction, model training, and classification.

You can create a simple diagram in Word (Insert → Shapes) and label it as:.

Fig. 1. Phishing Email Detection System Architecture

Suggested flow: Email Input → Preprocessing → Feature Extraction → Model → Classification Output

Figures are placed either at the top or bottom of the column and are referenced as "Fig. 1" in the text.

c) Implementation Details and Reproducibility

The implementation of the phishing detection system is carried out using Python and standard machine learning libraries. The complete codebase, including preprocessing scripts, feature extraction methods, and trained models, is made available through a public GitHub repository.

Providing access to the implementation ensures reproducibility and allows other researchers to validate and extend the work. The system is designed to be modular so that additional features or models can be integrated easily.

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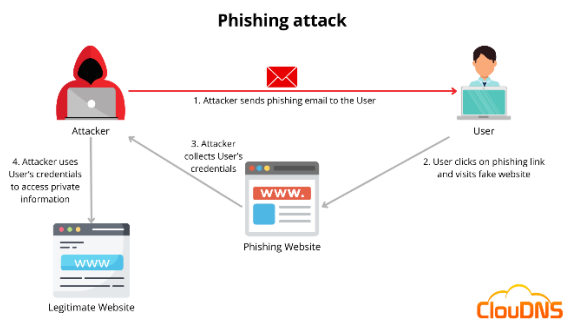
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